

Manuel Cattoni

[Portfolio](#) | [Github](#) | [Linkedin](#)

Varese, Italy | manuel.cattoni93@gmail.com

In Berlin for a German language course until Nov. 21, 2026; available full-time (100%) Dec. 2026 to mid-Feb. 2027, then 60% (Master starts)

PROFESSIONAL SUMMARY

Data Science and AI graduate (B.S., SUPSI) with practical experience in multimodal time-series pipelines, machine learning validation, and full-stack data dashboards, complemented by independent web, mobile, and applied-ML projects. Experienced in Linux environments, local server infrastructure, and customer operations. Planning to start a part-time MSE (Data Science profile, with electives in Business and Biomedical Engineering) in February 2027; eager to bring that same versatility and hands-on curiosity to real client work in the meantime.

EDUCATION

SUPSI

B.S. in Data Science and Artificial Intelligence (Graduated)

I.S.I.S. J.M. Keynes

High School Diploma in Computer Science

Lugano, Switzerland

Sept. 2023 – Sept. 2026

Gazzada Schianno, Italy

Sept. 2018 – July 2023

EXPERIENCE

Thesis Researcher & Data Annotator

MeDiTech (SUPSI) in collaboration with EOC

- Investigated 59 hours of 100 Hz 9-DoF IMU signals (accelerometer, gyroscope, magnetometer) from smart toys across 26 toddlers for ASD (Autism Spectrum Disorder) behavioral markers in the AUTOPLAY clinical project.
- Curated and standardized 4,800+ action intervals across 36 motor classes; recovered undocumented sensor-to-toy mappings using angular velocity contrast ratios.
- Engineered a session representation (*Behavioural Timelines*, 100 ms grid) and evaluated ML regressors/classifiers (LightGBM, Random Forest, MiniRocket) to predict ADOS-2 scores from play sessions, under nested leave-one-subject-out CV to eliminate leakage.
- Developed an offline, GDPR-compliant clinical dashboard using Dash, Plotly, and Flask to synchronize multi-angle video playback with sensor channels and timeline annotations for clinicians.

May 2026 – Sept. 2026

Lugano & Bellinzona, Switzerland

Undergraduate Peer Tutor

SUPSI

- Provided individual academic tutoring and technical problem-solving support in mathematics and computer science.
- Coordinated student appointments and delivered approximately 3 hours of weekly instructional sessions.

Nov. 2024 – Present

Lugano, Switzerland

Delivery Driver & Cashier

Storica Pizzeria della Stazione

- Worked at least 3 evening shifts per week alongside full-time university studies.

Dec. 2023 – Feb. 2026

Malnate, Italy

Data Entry Clerk

BluRigena

- Entered and verified business records to support administrative workflows.

Dec. 2024 – Jan. 2025

Angera, Italy

IT Internships (High School)

EasyCore, inVar, iMania

- Configured on-premises servers and hardware inventory, deployed IP surveillance cameras, and maintained a company website via WordPress.

Feb. 2021 – Nov. 2022

Varese area, Italy

SELECTED PERSONAL PROJECTS ([PORTFOLIO](#))

Polify (Next.js, React, TypeScript, Tailwind CSS): Political affinity test that explains each of 68 policy topics before the user answers, then computes alignment with parties. To be finalised and published in time for the 2027 Italian elections.

PantryPilot (Flutter, Node.js, Gemini API): Helps people living alone manage diet, pantry, and groceries: reads a diet plan from a photo or PDF, builds a priced shopping list, and suggests recipes to avoid waste.

TECHNICAL SKILLS & INTERESTS

Programming & ML: Python (NumPy, SciPy, Pandas, Scikit-learn, LightGBM, PyTorch), R, SQL, C, C++, C#

Tools & Frameworks: Docker, Git, Linux, Dash/Plotly, Flask, SLURM, WordPress; from personal projects: Java, React, Next.js, Node.js, JavaScript/TypeScript, Tailwind CSS, Flutter

Methodologies: Deep Learning (Autoencoders, GNNs), Bayesian Optimization & Gaussian Processes, Reinforcement Learning, Anomaly Detection, Clustering & Feature Selection, Time-Series Analysis (Wavelets, Welch PSD, MiniRocket), Nested Cross-Validation, SHAP

Languages: Italian (Native), English (B2), German (A1)

Interests: Motorsport, Finance, Football, Fitness